

Shrey Patel

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Technical Skills

Languages: Java, Python, TypeScript/JavaScript, C/C++, C#, SQL

Frameworks & Libraries: Spring, .NET, Flask, React, Next.js, Node.js, LangChain, JUnit, Cypress, Jest

Messaging & Databases: RabbitMQ, Kafka, PostgreSQL, MongoDB, Redis

Cloud/DevOps: AWS, Docker, Kubernetes, Jenkins, Terraform, Git

Experience

Full-Stack Software Engineer

Oct 2025 – Present

Integration Innovation

Huntsville, AL

- Shipped a Command and Control (C2) platform to production, becoming the default control path across 13 autonomous vehicles and the primary feature demoed to the US Marine Corps.
- Built a geospatial mission-planning toolset on OpenLayers from scratch, enabling operators to define geofences, program search areas, and route autonomous vehicles directly on a live map.
- Designed and built a real-time operator handoff system in .NET enabling live ownership transfers across vehicle control queues, with atomic state transitions guaranteeing exclusive control with zero conflicts across 50+ concurrent users.
- Built an end-to-end test suite covering 23 mission-critical flows with Cypress and Cucumber, cutting release QA from 2 days to 20 minutes.

Full-Stack Software Engineer Intern

Aug 2024 – Feb 2025

Peraton

Blacksburg, VA

- Engineered high-throughput Java microservices to orchestrate real-time NASA satellite telemetry streams across 5 distributed ground stations, ensuring zero data loss for 100+ concurrent flight controllers.
- Redesigned RabbitMQ message topology for ground station subsystems, reducing queuing latency 70% (800 ms → 240 ms) and eliminating message loss under peak operator load.
- Built an internal RAG agent over ops documentation, deflecting 55% of tier-1 questions and rolling out to the full team.

Full-Stack Software Engineer Intern

May 2024 – Aug 2024

CACI

Reston, VA

- Re-architected an enterprise data validation engine across 8M+ records, leveraging composite indexing to cut query latency by 80% (from 1s to under 200 ms).
- Built internal analytics APIs serving thousands of daily reports, replacing a \$30K/year third-party vendor tool.

Full-Stack Software Engineer Intern

May 2023 – Aug 2023

Ventura

Reston, VA

- Scaled Node.js APIs serving 1,000+ daily users by migrating high-traffic endpoints to a Redis cache-aside pattern, cutting TTFB 90% and increasing throughput 3x.
- Identified over-provisioned Kubernetes workloads through a Jenkins-integrated performance testing suite, reducing CPU usage 25% and memory and pod counts 15%.

Projects

Personal Portfolio Website | *Next.js, LangChain, OpenAI, Vercel AI SDK, Redis*

Aug 2025 – Present

- Implemented a RAG pipeline using LangChain and GPT-5 to answer natural language queries about my experience, routing semantic search through a vector store via Vercel's AI SDK.
- Added a serverless Redis caching layer for repeated queries, reducing inference latency and cutting operational cost.

Jobhub.cs.vt.edu | *React, Flask, PostgreSQL, Docker, Kubernetes*

Capstone Project

- Designed and shipped an university industry-partnership portal for Virginia Tech's CS department as primary backend developer on a 5-person team, serving 100+ industry partners.
- Delivered a multi-tier sponsorship system in Flask and PostgreSQL managing partner relationships, event workflows, and financial tiers, replacing a manual email-based coordination process used by the department.

Education

Bachelor of Science, Computer Science

May 2025

Virginia Tech

GPA: 3.9

Relevant Courses: Data Structures and Algorithms, Operating Systems, Computer Architecture, Database Management Systems, Cloud Software Development, Intro to AI, Machine Learning